Softmax的第i个分量为 $\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$

计j, $J_{i,j} = \frac{\partial \hat{y}_i}{\partial z_j} = \frac{-e^{z_i} e^{z_j}}{(\sum_{k=1}^{K} e^{z_k})^2}$

$i=j$, $J_{i,i} = \frac{\partial \hat{y}_i}{\partial z_i} = \frac{-e^{z_i} \cdot e^{z_i}}{(\sum_{k=1}^{K} e^{z_k})^2} + \frac{e^{z_i}}{\sum_{k=1}^{K} e^{z_k}} = \frac{\sum_{k=1, k\neq i}^{K} e^{z_i} e^{z_k}}{(\sum_{k=1}^{K} e^{z_k})^2}$

记 $S = \sum_{k=1}^{K} e^{z_k}$

则 $J_{i,j} = \begin{cases} -e^{z_i} e^{z_j} / S^2 & i \neq j \\ e^{z_i}(S - e^{z_i}) / S^2 & i = j \end{cases}$

所以 ☆ Softmax的Jacobi矩阵为

$$-\frac{1}{S^2}\begin{bmatrix} e^{z_1}(e^{z_1} - S) & e^{z_1} e^{z_2} & \cdots & e^{z_1} e^{z_K} \\ e^{z_2} e^{z_1} & e^{z_2}(e^{z_2} - S) & \cdots & e^{z_2} e^{z_K} \\ \cdots & & & \cdots \\ e^{z_K} e^{z_1} & e^{z_K} e^{z_2} & \cdots & e^{z_K}(e^{z_K} - S) \end{bmatrix}$$

单样本 $l = -\sum_{i=1}^{K} y_i \log \hat{y}_i$

$$\frac{\partial l}{\partial z_j} = \sum_i \frac{\partial l}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial z_j}$$

$$= \sum_{i \neq j} \frac{y_i}{\hat{y}_i} \frac{e^{z_i} e^{z_j}}{S^2} + \frac{y_j}{\hat{y}_j} \cdot \frac{e^{z_j}(e^{z_j} - S)}{S^2}$$

$$= \frac{e^{z_j}}{S} - y_j$$

$$= \hat{y}_j - y_j$$

而 $L = \frac{1}{m} \sum_{i=1}^{m} l^{(i)}$ ~~$= \frac{1}{m}(y_j$~~

$\frac{\partial L}{\partial z_j}$ 因此小批量平均 $\frac{\partial L}{\partial z_j^{(i)}} = \frac{1}{m}(\hat{y}_j^{(i)} - y_j^{(i)})$

对共享参数，是将各样本的梯度贡献相加后取平均